

FOR THE TREATMENT OF DRY EYE

**DIQUAS<sup>®</sup> ophthalmic solution 3%**

(Diquafosol sodium 3%)

75479404



004681



**Description**

DIQUAS<sup>®</sup> ophthalmic solution is a clear, colorless and sterile aqueous ophthalmic solution. Each mL contains 30 mg of diquafosol sodium. It also contains dibasic sodium phosphate hydrate, disodium edetate hydrate, sodium chloride, potassium chloride, chlorhexidine gluconate, and dilute hydrochloric acid or sodium hydroxide as pH adjuster. Its pH is 7.2 - 7.8 and its osmolar ratio is 1.0 - 1.1.

**Indications**

The product should be used in patients diagnosed with dry eye, associated with keratoconjunctival epithelium disorders that accompany lacrimal fluid abnormality.

**Dosage and Administration**

Instill 1 drop a time in the affected eye(s) 6 times daily.

**Contraindications**

Patients with a history of hypersensitivity to any of the ingredients of this product.

**Precautions**

**1.Adverse Reactions**

Adverse drug reactions (including abnormal changes in laboratory test values) were reported in 155 of 655 patients (23.7%) in clinical trials conducted in Japan before approval. The major adverse reactions were eye irritation in 44 patients (6.7%), eye discharge in 31 patients (4.7%), conjunctival hyperaemia in 24 patients (3.7%), eye pain in 18 patients (2.7%), eye itching in 16 patients (2.4%), foreign body sensation in eyes in 14 patients (2.1%) and ocular discomfort in 7 patients (1.1%), etc.

Adverse reactions were reported in 202 of 3,196 patients (6.3%) in post marketing observational study in Japan. The major adverse reactions were eye irritation in 30 patients (0.9%), eye discharge in 30 patients (0.9%), eye pain in 22 patients (0.7%), lacrimation increased in 20 patients (0.6%) and blepharitis in 19 patients (0.6%), etc.

If an adverse drug reaction is observed, appropriate measures including discontinuing administration should be taken.

| Type             | Incidence                                                                                                          |                |                                                                                                                                                                                                                                                                          |
|------------------|--------------------------------------------------------------------------------------------------------------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  | Incidence unknown                                                                                                  | ≥ 5%           | 0.1 - < 5%                                                                                                                                                                                                                                                               |
| Hypersensitivity | —                                                                                                                  | —              | Blepharitis                                                                                                                                                                                                                                                              |
| Ophthalmic       | Corneal epithelium disorder (filamentary keratitis, keratitis superficial, corneal erosion, etc.) , conjunctivitis | Eye irritation | Eye discharge, conjunctival hyperaemia, eye pain, eye itching, foreign body sensation in eyes, visual discomfort, hyposphagma, abnormal sensation in eye (eyes dry feeling of, eye strange sensation of, sticky eye sensation), vision blurred, photophobia, lacrimation |
| Others           | —                                                                                                                  | —              | Headache, increased eosinophils, elevated ALT(GPT)                                                                                                                                                                                                                       |

Incidence was calculated based on the clinical study results up to approval.

**2.Pediatric Use**

The safety of this product to low birth weight infants, neonates, infants or children has not been established. (No clinical experience.)

**3.Precautions concerning Use**

- 1) Route of administration: Ophthalmic use only.
- 2) At the time of administration:
  - (1) Instruct the patient to be careful not to touch the tip of the bottle to the eye directly in order to avoid the contamination of the drug.
  - (2) When more than one ophthalmic solution is used, at least 5 minutes of intervals should be taken.

**4.Pregnancy and Lactation**

There are no adequate data for the use of diquafosol in pregnant women. Animal studies do not indicate direct or indirect harmful effects with respect to reproductive toxicity. It is unknown whether diquafosol and/or its metabolites are excreted in human milk.

**5.Interaction with other medicaments**

No interaction studies have been performed with diquafosol.

**6.Overdose and treatment**

Overdose is unlikely to occur after ocular administration. If overdose occurs, treatment should be symptomatic.

**7.Warning and Precaution**

DIQUAS contains chlorhexidine. Chlorhexidine is known to induce hypersensitivity, including generalized allergic reactions and anaphylactic shock. The prevalence of chlorhexidine hypersensitivity is unknown, but available literature suggests this is likely to be very rare. DIQUAS should not be administered to anyone with a possible history of an allergic reaction to chlorhexidine.

If any signs or symptoms of a suspected hypersensitivity reaction such as itching, skin rash, redness, swelling, breathing difficulties, light headedness, and rapid heart rate develop, immediately stop using the product. Appropriate therapeutic countermeasures must be instituted as clinically indicated.

**8.Undesirable Effects**

Immune system disorders  
Frequency not known: Hypersensitivity including anaphylactic shock

**Pharmacodynamics/Pharmacokinetics**

**1.Pharmacodynamics**

**1) Mechanism of action**

Diquafosol sodium is a mucin/aqueous secretagogue acting on the P2Y<sub>2</sub> receptors on conjunctival epithelium and goblet cell membranes. The increased intracellular calcium ions promotes water and mucin secretion.

**2) Stimulatory action on secretion of tear fluid including mucin**

- (1) A single dose administration of diquafosol sodium into the eyes of normal animals (rabbits and rats) promoted tear fluid secretion and mucin secretion from the conjunctival cells.
- (2) A single dose administration of diquafosol sodium into the eyes of dry eye model rats promoted tear fluid secretion. Repeated dose administration increased mucin contents in the conjunctival tissues.

**3) Stimulatory action on mucin production in corneal epithelial cells**

Diquafosol sodium stimulated gene expression and protein production of membrane-associated mucin in corneal epithelial cells.

**4) Improvement of corneal epithelial damage**

Repeated dose administration of diquafosol sodium 6 times daily for 4 weeks improved corneal epithelial damage in rat dry eye model in a dose-dependent manner, and exhibited the maximal effect at the concentration of 1% or higher. Repeated dose of 1% diquafosol sodium for 2 weeks exhibited the maximal improvement effect when daily administration exceeded 6 times.

2.Pharmacokinetics

1) Plasma concentrations

After topical administration of diquafosol sodium solution either at concentrations of 0.3%, 1%, 3% or 5% to the eye of healthy adult volunteers one drop, once daily for one day, 6 times daily for one day or 6 times daily for 7 days, the plasma concentrations of diquafosol sodium and its metabolites were measured. Those of diquafosol sodium were below the lower limit of quantitation (2ng/mL) at every time point in all of the volunteers. Its metabolites (UTP, UDP, UMP and uridine) did not affect physiological concentrations derived from the endogenous components. (Note: The approved concentration of this product is 3%.)

2) Ocular tissue distribution in animals

(Rabbits)

Following a single topical administration of 3% <sup>14</sup>C-diquafosol sodium ophthalmic solution to rabbit eyes, the radioactivity was distributed in the extraocular tissues including the conjunctiva and cornea, and reached the maximum radioactive concentrations in the cornea and conjunctiva at 5 minutes after administration. Thereafter, the radioactivity reduced to 4 to 30% of the maximum concentrations at 24 hours after administration.

3) Metabolism

(Human *in vitro*)

*In vitro* metabolism reaction using human plasma and human liver microsome demonstrated that diquafosol sodium was rapidly metabolized, and UMP, uridine and uracil were produced.

(Rabbits)

At 30 minutes after instillation of 3% <sup>14</sup>C-diquafosol sodium ophthalmic solution to rabbit eyes, diquafosol sodium was hardly detected in the ocular tissues, and instead UTP, UDP, UMP, uridine and uracil were detected.

CLINICAL STUDIES

1.Late phase II study (Randomized, double-masked comparative study using a placebo ophthalmic solution as a control drug)

In the late phase II study in patients with dry eye, DIQUAS (in 93 patients including 16 patients with Sjögren’s syndrome) significantly lowered fluorescein staining scores\* in the cornea and rose bengal staining scores\* in the cornea and conjunctiva compared to a placebo ophthalmic solution (as a control drug in 93 patients including 23 patients with Sjögren’s syndrome)

Comparison of Fluorescein Staining Scores\*

|                                                                  | DIQUAS (n=93)          | Placebo (n=93) |
|------------------------------------------------------------------|------------------------|----------------|
| Amount of change (Week 4 or at discontinuation)                  | -1.55±0.13             | -0.95±0.14     |
| Intergroup difference [95% Confidence Interval (DIQUAS-Placebo)] | -0.60 [-0.98 to -0.22] |                |

(mean ± standard deviation)

Comparison of Rose Bengal Staining Scores\*

|                                                                  | DIQUAS (n=93)          | Placebo (n=93) |
|------------------------------------------------------------------|------------------------|----------------|
| Amount of change (Week 4 or at discontinuation)                  | -1.71±0.23             | -0.86±0.21     |
| Intergroup difference [95% Confidence Interval (DIQUAS-Placebo)] | -0.85 [-1.46 to -0.24] |                |

(mean ± standard deviation)

2.Phase III study (Randomized, double-masked comparative study using 0.1% purified sodium hyaluronate ophthalmic solution as a control drug)

In the phase III study in patients with dry eye, DIQUAS (in 144 patients including 36 patients with Sjögren’s syndrome) equivalently lowered fluorescein staining scores\* in the cornea and significantly lowered rose bengal staining scores\* in the cornea and conjunctiva compared to a 0.1% purified sodium hyaluronate ophthalmic solution (as a control drug in 142 patients including 32 patients with Sjögren’s syndrome)

Comparison of Fluorescein Staining Scores\*

|                                                                 | DIQUAS (n=144)          | 0.1% HA (n=142) |
|-----------------------------------------------------------------|-------------------------|-----------------|
| Amount of change (Week 4 or at discontinuation)                 | -2.12±0.14              | -2.08±0.13      |
| Intergroup difference [95% Confidence Interval (DIQUAS-0.1%HA)] | -0.03 [-0.405 to 0.338] |                 |

0.1% HA: 0.1% purified sodium hyaluronate ophthalmic solution (mean ± standard deviation)

Comparison of Rose Bengal Staining Scores\*

|                                                                 | DIQUAS (n=144)         | 0.1% HA (n=141) |
|-----------------------------------------------------------------|------------------------|-----------------|
| Amount of change (Week 4 or at discontinuation)                 | -3.06±0.19             | -2.38±0.18      |
| Intergroup difference [95% Confidence Interval (DIQUAS-0.1%HA)] | -0.67 [-1.18 to -0.16] |                 |

0.1% HA: 0.1% purified sodium, hyaluronate ophthalmic solution (mean ± standard deviation)

3.Long-term administration study (Phase III)

In a long-term administration study in patients with dry eye (in 244 patients including 9 patients with Sjögren’s syndrome and 2 patients with Stevens-Johnson syndrome), the group under DIQUAS administration exhibited lower fluorescein staining scores\* in the cornea and rose bengal staining scores\* in the cornea and conjunctiva throughout the 52 weeks administration period, and longer continued effect compared to those before administration.

\* Scoring method in the clinical studies

In the fluorescein staining, severity of a corneal damage was graded on a nine-point scale by scoring the fluorescein staining intensity at 3 sites (superior, middle and inferior) which were individually graded from 0 to 3 points. In the rose bengal staining, severity of a corneal and conjunctival damage was graded on a 15-point scale by scoring the rose bengal staining intensity at 5 sites (superior, middle and inferior sites of the cornea, and nasal and temporal conjunctiva) which were individually graded from 0 to 3 points.

**Storage** Store below 30°C. After opening, to be used within one month.

**How supplied** Plastic bottles of 5 mL

Manufactured by  
**SANTEN PHARMACEUTICAL CO., LTD.**  
Noto plant: 2-14, Shikunami, Hodatsushimizu-cho, Hakui-gun, Ishikawa, Japan.

Product Registration Holder:  
**SANTEN PHARMA MALAYSIA SDN.BHD.**  
Unit #23A-10, Q Sentral,  
No. 2A, Jalan Stesen Sentral 2, Kuala Lumpur Sentral,  
50470 Kuala Lumpur, Malaysia.

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